Programming Fundamentals – COC1071
Lab Plan – 03

Learning Objective:

- *Selection*
- *Multiple Selection*

NOTE: For each question, first make algorithm and Flow-chart then write code according to algorithm or flow-chart.

1. Write a program to calculate the electricity bill based on usage in kilowatt-hours (kWh). The rates are as follows:
- For the first 100 kWh: $0.50 per kWh
- For the next 200 kWh (101-300): $0.75 per kWh
- For usage above 300 kWh: $1.20 per kWh

If the usage is more than 500 kWh, an additional surcharge of 10% is applied to the total bill.

2. Write a program that simulates an ATM machine. The program should ask the user to input the withdrawal amount and check if the withdrawal is possible based on available balance and the following conditions:
- Minimum withdrawal amount is $20.
- The withdrawal amount must be a multiple of $10.
- The account balance should be sufficient for the withdrawal.

3. Write a program that calculates the total price of an item after applying a discount. The discount depends on the original price:
- Price less than $100: 5% discount
- Price between $100 and $500: 10% discount
- Price more than $500: 20% discount

4. Write a program that calculates the parking fee based on the number of hours a car is parked:
- Up to 2 hours: $5
- 3 to 5 hours: $10
- More than 5 hours: $15

5. Write a program that calculates whether a user has met their daily step goal. The daily goal is 10,000 steps. The program should ask the user for the number of steps they've taken and print one of the following:
- "Goal achieved!" if steps >= 10,000
- "Almost there!" if steps between 8,000 and 9,999
- "Keep going!" if steps < 8,000

6. Write a program that takes the current temperature as input and provides recommendations:
    - If the temperature is below 0°C: "It's freezing outside!"
    - If the temperature is between 0°C and 20°C: "Wear a jacket."
    - If the temperature is above 20°C: "It's a warm day."

7. Write a program that calculates a student's final grade in a course based on different categories:
    - Assignments: 40% of the final grade
    - Quizzes: 20% of the final grade
    - Exams: 40% of the final grade

    The program should accept grades for each category and calculate the weighted average. Classify the final grade into A, B, C, D, or F.

8. Write a program that simulates a hotel room reservation system. The user can choose from different room types:
    - Single room: $100 per night
    - Double room: $150 per night
    - Suite: $300 per night

    The program should apply a discount:
    - 10% discount if the stay is longer than 7 days
    - 15% discount if the stay is longer than 14 days

9. Write a program that simulates an event ticket booking system. The user can book tickets for different types of seats:
    - Standard: $50
    - Premium: $100
    - VIP: $200

    Apply group discounts:
    - 10% discount for booking more than 5 tickets
    - 15% discount for booking more than 10 tickets

10. Write a program to manage exam results for students. The system should allow:

    - Adding exam marks for a student (for multiple subjects)
    - Calculating the total and average marks for each student
    - Generating a report card (pass/fail based on an average passing mark of 50)